

***FACET*-NLO: Factorization Agent for Collider and Event-shape Theory**

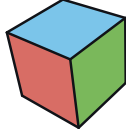
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***FACET*-NLO**

ABSTRACT: This note records the current *FACET*-NLO workflow for exact analytic hard coefficients in factorized collider observables. A process card determines an ordered amplitude–conjugate-amplitude pair, leading-twist collinear density projectors, phase space, and physical cuts. The resulting numerator and literal denominators are partial fractioned before integral families are chosen and are then reduced by cut-aware integration-by-parts identities. The central new stage is a constrained master-integral agent. It receives a frozen post-IBP manifest, constructs an exact physical parameterization, evaluates compact parametric densities with *SUBTROPICA*, proves changes of variables and branch conditions, and accepts output only after exact identities and independent checks pass. Fixed-point *AMFLOW* values and numerical quadrature are validation data, not the analytic result. At NLO the present implementation evaluates six physical master instances symbolically in the invariants and dimensional regulator. At NNLO, three difficult multiscale families containing 17 masters have now been solved analytically through ϵ^2 and checked coefficient by coefficient against independent *AMFLOW* values. A separate structural sweep reduces all 68 denominator-class representatives of the saved double-real process to 4,242 active family labels and 157 canonical support shapes. This exhausts the denominator representatives but does not yet prove completeness over every numerator in all 666 diagram pairs. Endpoint factors and plus distributions remain explicit through final reconstruction, and *BMHV* evanescent information is retained for polarized channels. The note separates the physics contracts from their implementation and records discarded or secondary approaches only in its final section.

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1 Scope and acceptance criterion

The immediate process is single-inclusive hadron production,

$$p(P_a) + p(P_b) \longrightarrow h(P_h) + X, \quad (1.1)$$

with unpolarized, longitudinally polarized, and transversely polarized hard parts. The reusable computational unit is one ordered diagram pair and one polarization. Different pairs may generate different partial fractions and integral families even when they belong to the same partonic channel.

The required output is an exact analytic function of the physical invariants, convolution variables, and dimensional regulator. In particular, the calculation must determine the Laurent coefficients in ϵ , the endpoint powers, and the coefficients of $\delta(1-w)$ and the plus distributions. Values at selected kinematic points are checks. They cannot replace the analytic result.

This document keeps two layers separate.

- Sections 2–7 define the mathematical objects and the evidence required to accept them.
- Sections 8–12 record the current software realization and its verified coverage.

The final Section 13 records alternative routes that were tested but are not the present production calculation.

Part I

Concepts and formulae

2 Collinear factorization and density projectors

At leading power, a single-inclusive cross section has the schematic form [1]

$$d\sigma = \sum_{a,b,c} \int_0^1 dx_a dx_b dz \mathcal{J}(x_a, x_b, z) f_{a/A}(x_a) f_{b/B}(x_b) D_{h/c}(z) d\hat{\sigma}_{ab \rightarrow cX}. \quad (2.1)$$

The Jacobian $\mathcal{J}$ and all conventional momentum-fraction powers are kept outside the hard coefficient. The collinear map is

$$k_a = x_a P_a, \quad k_b = x_b P_b, \quad P_h = z k_c. \quad (2.2)$$

After this map and the explicit convolution measure are removed, the hard coefficient may depend on the partonic invariants but must not contain an additional nontrivial dependence on x_a , x_b , or z .

For a quark leg, the leading-twist density matrix is

$$\Phi_q(x) = \frac{1}{2} [f_1(x) \not{x} + \lambda g_{1L}(x) \gamma_5 \not{x} + h_1(x) i \sigma^{nS_T} \gamma_5], \quad (2.3)$$

and the fragmentation matrix is

$$\Delta_q(z) = \frac{1}{2} [D_1(z) \not{p}_h + \lambda_h G_{1L}(z) \gamma_5 \not{p}_h + H_1(z) i \sigma^{n_h S_{hT}} \gamma_5]. \quad (2.4)$$

Here $i \sigma^{ab} = -[\not{a}, \not{b}]/2$. The antiquark convention reverses the helicity term while preserving the unpolarized and transversity terms. These are the standard leading-twist correlator structures [2].

The front end first performs ordinary spin and polarization sums. It then replaces spin sums associated with observed hadrons by Eqs. (2.3) and (2.4). This ordering is important: the generated amplitude is not edited by recognizing isolated spinor numerators or guessed momentum signs.

3 Cut integrals and invariant reduction

For a one-particle real emission, eliminate one final-state momentum using overall momentum conservation. A typical remaining integral is

$$\int d^D k \, \delta_+(D_1) \delta_+(D_2) \frac{N(k)}{D_3^{a_3} \cdots D_N^{a_N}}, \quad D = 4 - 2\epsilon. \quad (3.1)$$

Every cut includes its positive-energy orientation. Reverse unitarity uses [3]

$$\text{Cut}_+^{(n)}(D) \equiv \frac{1}{2\pi i} [(D - i0)^{-n} - (D + i0)^{-n}] = \frac{(-1)^{n-1}}{(n-1)!} \delta_+^{(n-1)}(D). \quad (3.2)$$

Thus a doubled cut is $-\delta'_+(D)$. It is neither a second cut line nor a derivative with respect to dimension.

A reduced master is identified by

$$M = (\mathcal{F}, \mathbf{a}, C, \boldsymbol{\xi}; \mathcal{S}), \quad (3.3)$$

where $\mathcal{F}$ is the denominator family, $\mathbf{a}$ its index vector, C the cut slots, $\boldsymbol{\xi}$ their energy orientations, and $\mathcal{S}$ the partition of integration momenta into phase-space, forward-virtual and conjugate-virtual sectors. The last datum is inferred from the process setup when a

boundary-value prescription is required; it is not copied as a sign label on every denominator. Dropping the cut data or the loop-sector partition can merge physically different integrals. The cut-family condition is

$$I_{\mathcal{F}}(\mathbf{a}) = 0 \quad \text{if any designated cut index is nonpositive.} \quad (3.4)$$

The literal denominators of a diagram need not be linearly independent. Generalized partial fractioning must therefore precede family construction [4]:

$$\boxed{\begin{array}{l} \text{literal denominators} \longrightarrow \text{cut-preserving fractions} \\ \longrightarrow \text{complete families} \longrightarrow \text{IBP reduction} \end{array}} \quad (3.5)$$

Every surviving fraction must retain every mandatory cut. Irreducible scalar products are appended only after fractioning, so the family spans the scalar products that actually occur rather than a process-specific guessed list.

3.1 Fixed-sign theorem for non-virtual propagators

The phase-space part of the reduction may generate ordinary denominators that contain no virtual loop momentum. Such a denominator can be used without an $i0$ label only after its sign has been proved on the complete physical cut support. For massless two-beam kinematics this question has an exact finite answer.

Let k_a and k_b be future-directed incoming null momenta and let $p_1, \dots, p_n$ be future-directed outgoing null momenta, with

$$k_a + k_b = \sum_{i=1}^n p_i, \quad s = 2k_a \cdot k_b > 0. \quad (3.6)$$

For real coefficients a, b, c_i , define

$$Q = ak_a + bk_b + \sum_{i=1}^n c_i p_i, \quad \mu_{ij} = (a + c_i)(b + c_j), \quad i \neq j. \quad (3.7)$$

Then on the closure of the physical massless phase space,

$$\boxed{\min_{i \neq j} \mu_{ij} \leq \frac{Q^2}{s} \leq \max_{i \neq j} \mu_{ij}} \quad (3.8)$$

and both bounds are sharp. Consequently, Q^2 has a fixed nonnegative sign if all $\mu_{ij} \geq 0$, and a fixed nonpositive sign if all $\mu_{ij} \leq 0$. A zero bound is allowed: it corresponds to a soft or collinear boundary. If the set $\{\mu_{ij}\}$ contains both signs, the denominator is not certified and the reduction stops.

To prove Eq. (3.8), introduce the nonnegative numbers

$$x_i = \frac{2k_b \cdot p_i}{s}, \quad y_i = \frac{2k_a \cdot p_i}{s}, \quad z_{ij} = \frac{2p_i \cdot p_j}{s} = z_{ji}. \quad (3.9)$$

They satisfy

$$\sum_i x_i = \sum_i y_i = 1, \quad x_i + y_i = \sum_{j \neq i} z_{ij}, \quad \sum_{i < j} z_{ij} = 1. \quad (3.10)$$

For a subset $A \subseteq \{1, \dots, n\}$, write $K_A = \sum_{i \in A} p_i$ and $z(A) = \sum_{i < j \in A} z_{ij} = K_A^2/s$. In beam light-cone coordinates, $k_b - K_A$ is the difference between one incoming beam and a subset of the physical final state, and

$$(k_b - K_A)^2 \leq 0. \quad (3.11)$$

Equivalently,

$$z(A) \leq \sum_{i \in A} x_i. \quad (3.12)$$

The analogous statement with k_a gives the corresponding inequality for y_i . Equation (3.12) is precisely the max-flow/min-cut condition for fractionally orienting the weighted complete graph with undirected edge weights z_{ij} and prescribed outgoing weights x_i . Hence there are numbers $\lambda_{ij} \geq 0$, $i \neq j$, such that

$$\lambda_{ij} + \lambda_{ji} = z_{ij}, \quad \sum_{j \neq i} \lambda_{ij} = x_i. \quad (3.13)$$

Using Eq. (3.10), their incoming weights are automatically

$$\sum_{i: i \neq j} \lambda_{ij} = y_j, \quad \sum_i \sum_{j \neq i} \lambda_{ij} = 1, \quad (3.14)$$

where the first and second sums in Eq. (3.14) have respectively fixed j and unrestricted ordered pairs. Expanding Q^2 and using these marginals gives the convex decomposition

$$\frac{Q^2}{s} = \sum_{i \neq j} \lambda_{ij} (a + c_i)(b + c_j) = \sum_{i \neq j} \lambda_{ij} \mu_{ij}. \quad (3.15)$$

Equation (3.8) follows immediately. Finally, the boundary configuration $p_i \rightarrow k_a$, $p_j \rightarrow k_b$, with all other final momenta soft, gives $Q^2/s \rightarrow \mu_{ij}$. Every corner value is therefore a physical boundary limit, proving that the bounds are sharp.

4 Cut-aware topology equivalence and family reuse

A scalar denominator family may occur in many ordered diagram pairs and with many choices of loop-momentum routing. Such repetitions should not require independent IBP reductions. For a cut integral, however, equality of the denominator polynomials is not sufficient. In particular, the replacement $q^\mu \rightarrow -q^\mu$ leaves a quadratic denominator $q^2 - m^2$ unchanged but reverses its positive- and negative-energy support. Likewise, an algebraic graph isomorphism does not determine whether a virtual loop belongs to the forward or conjugate side, nor whether a line has a distinguished physical role in the measurement.

The relevant equivalence relation must therefore act on decorated cut integrals rather than on bare denominator graphs.

We represent a topology by the record

$$\mathcal{T} = (\mathbf{D}, \boldsymbol{\ell}, \mathbf{p}, \mathcal{K}; C, \boldsymbol{\xi}, \mathcal{S}, \mathbf{m}), \quad \mathbf{D} = (D_1, \dots, D_N), \quad (4.1)$$

The ordered list $\mathbf{D}$ contains the generalized propagators, while $\boldsymbol{\ell} = (\ell_1, \dots, \ell_L)$ is the loop-momentum basis. $\mathbf{p}$ denotes the external momenta, and $\mathcal{K}$ is the set of exact external kinematic relations. The subset $C \subseteq \{1, \dots, N\}$ specifies the cut slots. For $i \in C$, the label $\xi_i = \pm 1$ specifies the energy orientation of the cut momentum q_i . The loop partition $\mathcal{S}$ assigns each integration momentum to the phase-space, forward-virtual or conjugate-virtual sector. The optional label $\mathbf{m}_i$ records line-level physical information, such as an identified final-state line, a measurement role, or an amplitude-conjugate-amplitude assignment. Masses and the algebraic type of each propagator are included in D_i , while cut orientations and loop sectors are retained separately.

At the level of on-shell support, the orientation label may be represented by

$$\Delta_{\xi_i}(q_i) \equiv \delta(q_i^2 - m_i^2) \theta(\xi_i q_i^0), \quad i \in C. \quad (4.2)$$

This notation is independent of the overall normalization chosen for the cut propagator in reverse unitarity.

Two records $\mathcal{T}$ and $\mathcal{T}'$ are cut-aware equivalent, $\mathcal{T} \simeq_{\text{cut}} \mathcal{T}'$, if there is an exact affine change of loop variables

$$\ell_r^\mu = \sum_{s=1}^L A_{rs} \ell_s'^\mu + \sum_a B_{ra} p_a^\mu, \quad A_{rs}, B_{ra} \in \mathbb{Q}, \quad |\det A| = 1, \quad (4.3)$$

and a permutation $\pi \in S_N$ such that

$$D_i(\ell, p) \Big|_{(4.3)} \stackrel{\mathcal{K}}{=} D'_{\pi(i)}(\ell', p), \quad i = 1, \dots, N. \quad (4.4)$$

The equality in Eq. (4.4) is strict: no slot-dependent multiplicative factor is allowed, and the mass and propagator type must agree. The determinant condition ensures that the dimensionally regulated integration measure is unchanged,

$$\prod_{r=1}^L d^d \ell_r = |\det A|^d \prod_{r=1}^L d^d \ell_r' = \prod_{r=1}^L d^d \ell_r', \quad d = 4 - 2\epsilon. \quad (4.5)$$

The permutation must preserve the cut subset,

$$\pi(C) = C'. \quad (4.6)$$

In addition, equality of the cut denominator polynomials must be supplemented by equality of their oriented on-shell support. For each $i \in C$, the mapped line momentum must satisfy

$$q_i(\ell, p) \Big|_{(4.3)} = s_i q'_{\pi(i)}(\ell', p), \quad s_i = \pm 1. \quad (4.7)$$

Indeed,

$$\begin{aligned}\Delta_{\xi_i}(s_i q'_{\pi(i)}) &= \delta(q_{\pi(i)}'^2 - m_i^2) \theta(s_i \xi_i q_{\pi(i)}'^0) \\ &= \Delta_{s_i \xi_i}(q'_{\pi(i)}),\end{aligned}\tag{4.8}$$

so preservation of the cut distribution requires

$$\boxed{\xi'_{\pi(i)} = s_i \xi_i}, \quad i \in C.\tag{4.9}$$

Thus a momentum reversal is allowed only when it is accompanied by the corresponding reversal of the stored energy orientation.

The affine map must preserve the loop-sector partition: a source integration momentum may mix only target momenta in the same member of $\mathcal{S}$. All available line-level physical labels must also be transported by the same permutation:

$$\mathbf{m}'_{\pi(i)} = \mathbf{m}_i \quad \text{whenever the labels are defined.}\tag{4.10}$$

Consequently, if the exponent in source slot i is a_i , the exponent in target slot $\pi(i)$ is also a_i . Equivalently,

$$a'_{\pi(i)} = a_i, \quad \text{or} \quad a'_j = a_{\pi^{-1}(j)}.\tag{4.11}$$

The induced integral identity is therefore

$$I_{\mathcal{T}}(a_1, \dots, a_N) = I_{\mathcal{T}'}(a_{\pi^{-1}(1)}, \dots, a_{\pi^{-1}(N)}).\tag{4.12}$$

This relation applies to integer auxiliary indices as well as positive propagator powers, subject to the no-pinching condition (3.4) on every designated cut slot.

For a collection of diagram-pair families, these conditions define a partition into cut-aware equivalence classes,

$$\{\mathcal{T}_\alpha\} = \bigsqcup_r [\mathcal{T}_r]_{\text{cut}}.\tag{4.13}$$

Only one representative $\mathcal{T}_r$ of each class must be submitted to an IBP reduction. For every other member, the affine loop-momentum map and the slot permutation π are retained, and the reduced integral is recovered by the index transport in Eq. (4.12). This is the precise sense in which equivalent topologies permit family reuse. A failed cut, orientation, loop-sector, or physical-label test places the candidate in a different class even when its undecorated denominator graph is isomorphic.

When the line-level physical labels $\mathbf{m}$ are absent, Eq. (4.12) certifies reuse of the cut-aware scalar IBP reduction. It does not, by itself, prove that two measured contributions belong to the same physical branch or carry the same measurement assignment.

As a simple two-loop example, consider the ordered propagator lists

$$\mathbf{D}_A = (\ell_1^2, \ell_2^2, (\ell_1 - \ell_2)^2, (\ell_1 - p)^2, (\ell_1 + \ell_2 - p)^2),\tag{4.14}$$

$$\mathbf{D}_B = (r_1^2, r_2^2, (r_1 - r_2)^2, (r_2 - p)^2, (r_1 + r_2 - p)^2).\tag{4.15}$$

The transformation

$$\ell_1 = r_2, \quad \ell_2 = r_1 \quad (4.16)$$

has $\det A = -1$ and induces the permutation $\pi = (12)$. The first two mapped line momenta acquire no additional sign: $q_{A,1} \rightarrow q_{B,2}$ and $q_{A,2} \rightarrow q_{B,1}$. With

$$C_A = C_B = \{1, 2\}, \quad (\xi_{A,1}, \xi_{A,2}) = (+1, -1), \quad (\xi_{B,1}, \xi_{B,2}) = (-1, +1), \quad (4.17)$$

and with matching loop sectors and physical labels in the remaining slots, the two records are equivalent. Their indices obey

$$I_A(a_1, a_2, a_3, a_4, a_5) = I_B(a_2, a_1, a_3, a_4, a_5). \quad (4.18)$$

Several nearby algebraic matches are not cut-aware equivalences. Keeping instead $(\xi_{B,1}, \xi_{B,2}) = (+1, -1)$ violates Eq. (4.9). A map that exchanges a forward virtual loop with a conjugate virtual loop is likewise rejected, even when all quadratic denominator polynomials match. Similarly, a map that sends a measured line to a line with a different physical role establishes at most a scalar-family relation. Finally, a proposed transformation such as $\ell_1 = 2r_1$, $\ell_2 = r_2$ has $|\det A| = 2$ and changes the loop measure by 2^d ; it therefore does not produce the strict index-only identity (4.12). These examples show why an algebraic family match must be followed by explicit cut-orientation, loop-sector, and physical tests.

Integration-by-parts identities follow from

$$0 = \int \prod_{r=1}^L d^D \ell_r \frac{\partial}{\partial \ell_i^\mu} \left[v^\mu \prod_j D_j^{-a_j} \right]. \quad (4.19)$$

They reduce the target integrals to a finite master set [5]. The implementation uses KIRA[6, 7]; the cut slots are part of the Kira family, not a filter applied to an ordinary reduction afterward.

5 The analytic master-integral agent

IBP reduction does not evaluate the masters. FACET-NLO treats master evaluation as a separate constrained task with a frozen input and an auditable output. The agent may propose a parameterization, split an integral, or recognize a special function, but none of those proposals is accepted without exact checks.

5.1 Post-IBP input contract

For each required master, the handoff contains

$$\mathcal{M}_{\text{in}} = \{\mathcal{F}, \mathbf{a}, C, \boldsymbol{\sigma}, c(\epsilon, \{s_i\}), \mathcal{R}, r_\epsilon, \mathcal{B}\}. \quad (5.1)$$

Here c is the exact IBP coefficient, $\mathcal{R}$ the physical region, r_ϵ the required Laurent depth, and $\mathcal{B}$ the BMHV or dimension shift sector. The manifest is immutable during evaluation.

In particular, a known final result is not allowed to alter the target list, normalization, or boundary data.

The manifest records several counts separately. These are physical master instances, family labels, active labels in one projected expression, support shapes after a proved variable change, and independent analytic functions after exact identities. The counts are not interchangeable.

5.2 Exact phase-space parameterization

The current analytic route converts a cut master to an ordinary integral over a physical domain only after IBP. Schematically,

$$M(\{s_i\}, \epsilon) = \mathcal{N}(\{s_i\}, \epsilon) \int_{\Omega_+} \prod_j dx_j \prod_r P_r(\mathbf{x})^{\alpha_r + \beta_r \epsilon}. \quad (5.2)$$

The map must be accompanied by four exact witnesses:

1. the inverse map and Jacobian;
2. equality between the original cut measure and Eq. (5.2);
3. positivity of every powered base P_r in Ω_+ ;
4. a phase-space volume or simpler-master normalization check.

Process-specific coordinates are permitted here because the invariant IBP problem is already complete. They are not prerequisites for partial fractioning or Kira.

Infinite intervals are mapped to positive projective variables, and exact rational inputs are retained. Machine numbers, unrestricted power expansion, or an unproved complex branch cause the task to fail.

5.3 Analytic integration and exact simplification

The compact density is evaluated with SUBTROPICA[8]. The desired output is an exact Laurent series whose coefficients are analytic functions or exact constants. A successful call must contain no unresolved integration wrapper, conditional expression, machine real, or indeterminate object.

The agent requests only the Laurent depth demanded by the exact IBP coefficient. Independent shapes may be evaluated concurrently, but each SUBTROPICA call is restricted to one kernel. Expression size alone is not a runtime predictor. If one density is too difficult, it may be split into smaller exact pieces only after proving the split identity, branch conditions, pole cancellation, and final recombination.

Special-function cleanup is performed after integration. Harmonic polylogarithm constants and multiple-zeta values are converted with explicit identities; a global simplifier is not allowed to choose branches silently.

5.4 Endpoint-safe coefficient jobs

A difficult master is decomposed into separately checkpointed Laurent coefficient jobs only after the endpoint regions have been resolved. The scientific checkpoint key is computed from the evaluated canonical payload, not from a held local symbol. In particular, a construction such as

```
Hash[HoldComplete[localCoefficient], "SHA256"]
```

is not reproducible inside a scoped module: it hashes the localized symbol name, whose serial number changes between kernels. FACET-NLO first materializes the coefficient data, assumptions, and requested order, and only then applies `HoldComplete` and the hash. Cross-kernel byte comparison of all serialized inputs is required before checkpoints can be reused.

Endpoint cancellation also fixes the unit of analytic integration. A raw GPL antiderivative can contain terms that diverge separately at an endpoint while their sum is finite. Such terms are combined and rewritten into individually convergent GPLs before endpoint evaluation. Exact residuals are checked first; high-precision positive-domain integration and an independent GPL evaluator then test the complete endpoint-safe sum. A structural classification of a coefficient job as linearly reducible is scheduling information, not an analytic result.

5.5 Exact relations and differential equations

Before counting independent masters, the agent tests scaling, symmetry, and IBP relations. It also tests dimension recurrences. For example, the representative doubled cut obeys

$$\frac{C_2}{C} = \frac{(1 - 2\epsilon)(2 - 3\epsilon)}{\epsilon W}. \quad (5.3)$$

This removes one independent analytic function even though C and C_2 are distinct integral instances.

When direct parametric integration does not finish, the analytic fallback is an exact differential equation,

$$\frac{\partial}{\partial x} \mathbf{M}(x, \epsilon) = A(x, \epsilon) \mathbf{M}(x, \epsilon) + \mathbf{b}(x, \epsilon), \quad (5.4)$$

with boundary data obtained analytically from a simpler cut master, an exact phase-space volume, a controlled soft or collinear limit, regularity, or a symmetry identity. Numerical samples may diagnose the equation but cannot be the final boundary condition.

If the system has been transformed to the canonical form

$$d\mathbf{g}(\mathbf{x}, \epsilon) = \epsilon \sum_r R_r d\log \phi_r(\mathbf{x}) \mathbf{g}(\mathbf{x}, \epsilon), \quad (5.5)$$

its transport is an ordered Chen series. FACET-NLO now delegates this exact noncommutative expansion to LIBRA [9]. This replaces manual enumeration of words while retaining the order of the residue matrices R_r . The package does not determine the physical boundary vector: region scalings, causal continuation, and normalization remain independent

analytic inputs. This distinction matters for the worked family in Sec. 6, whose boundary contains three nonuniform region modes. Applying a generic regular-point series solver directly at that singular boundary produces unresolved inverse blocks and is rejected rather than used as a boundary prescription.

5.6 Acceptance hierarchy

The accepted output is an association containing the exact result, Laurent coverage, assumptions, branch witnesses, source hashes, and check residuals. The validation hierarchy is

1. exact transformation, normalization, and algebraic residuals;
2. exact differential equations and boundary values;
3. exact scaling, symmetry, and cut-power identities;
4. independent high-precision AMFLOW values at physical points;
5. independent numerical quadrature for difficult pieces;
6. a fresh rerun with byte-identical canonical output.

Items 4 and 5 validate an analytic formula but never define it.

6 Analytic evaluation of an NNLO three-particle cut master

As a worked analytic example, we evaluate a genuine two-loop, three-particle cut integral with two independent dimensionless invariants. Its integration-by-parts reduction requires eight master integrals, its highest sector is coupled, and its physical boundary contains several nonuniform regions. The result below is an exact Laurent function of the kinematics and of the dimensional regulator. It provides one nontrivial NNLO stress test, rather than a claim that arbitrary NNLO families admit the same treatment.

6.1 Physical cut integral and kinematics

Let the massless external momenta satisfy

$$k_a^2 = k_b^2 = k_c^2 = 0, \quad 2k_a \cdot k_b = s, \quad -2k_a \cdot k_c = t, \quad -2k_b \cdot k_c = u, \quad (6.1)$$

with $D = 4 - 2\epsilon$. We define

$$P = k_a + k_b - k_c, \quad x = -\frac{t}{s}, \quad y = -\frac{u}{s}, \quad \Delta = 1 - x - y = \frac{P^2}{s}. \quad (6.2)$$

The analytic solution is constructed in the physical chamber

$$0 < y < (1 - x)^2 < 1 - x < 1, \quad (6.3)$$

which in particular implies $s > 0$, $P^2 > 0$, and $0 < x < 1$.

The master to be evaluated is

$$G_{\text{top}}(s, t, u; \epsilon) = \int d^D k_e d^D k_f \frac{\delta_+(k_e^2) \delta_+(k_f^2) \delta_+((P - k_e - k_f)^2)}{(P - k_e)^2 (P - k_f)^2 (k_a - k_c - k_f)^2 (k_a - k_e)^2}. \quad (6.4)$$

The common raw loop and phase-space normalization used elsewhere in the manuscript is understood. It cancels from the normalized result below. The three on-shell constraints carry positive-energy support. In reverse unitarity [3] we use

$$\text{Cut}_+^{(n)}(D) = \frac{1}{2\pi i} [(D - i0)^{-n} - (D + i0)^{-n}] = \frac{(-1)^{n-1}}{(n-1)!} \delta_+^{(n-1)}(D), \quad (6.5)$$

so a power two in a cut slot denotes the differentiated distribution $-\delta'_+(D)$, rather than an additional physical cut. The inherited causal prescriptions of ordinary propagators are left implicit in what follows.

6.2 Integral family, IBP closure, and genuine coupling

To differentiate and reduce Eq. (6.4), we embed it in the nine-slot family

$$\begin{aligned} D_1 &= k_e^2, & D_2 &= k_f^2, & D_3 &= (P - k_e - k_f)^2, \\ D_4 &= k_a \cdot k_f, & D_5 &= k_b \cdot k_e, & D_6 &= (P - k_e)^2, \\ D_7 &= (P - k_f)^2, & D_8 &= (k_a - k_c - k_f)^2, & D_9 &= (k_a - k_e)^2. \end{aligned} \quad (6.6)$$

The first three entries are cut slots, whereas D_4 and D_5 are irreducible scalar products needed to complete the two-loop scalar-product basis. We define

$$I_\nu = \int d^D k_e d^D k_f \prod_{j=1}^3 \text{Cut}_+^{(\nu_j)}(D_j) \prod_{j=4}^9 D_j^{-\nu_j}, \quad (6.7)$$

with the cut-family condition that $I_\nu = 0$ whenever any of ν_1, ν_2, ν_3 is nonpositive. Cut-aware integration-by-parts identities [5] close on the eight masters

$$\begin{aligned} G_1 &= I_{111000000}, & G_2 &= I_{111000010}, & G_3 &= I_{111000011}, & G_4 &= I_{112000011}, \\ G_5 &= I_{111001010}, & G_6 &= I_{111001011}, & G_7 &= I_{111001111}, & G_8 &= I_{112001111}. \end{aligned} \quad (6.8)$$

Here $G_7 = G_{\text{top}}$. The master G_8 has the same ordinary propagators but a differentiated third cut.

This is not a collection of eight independent one-scale integrals. Invariant differentiation mixes G_7 and G_8 , and lower sectors enter their inhomogeneous terms. Before canonicalization, the highest sector is a genuinely coupled 2×2 block. Moreover, its differential matrix contains a rank-one double pole on the physical singular surface $\Delta = 0$. The need to retain both the coupled top sector and the differentiated cut is what makes this a useful analytic NNLO example.

6.3 Flat differential system and canonical form

Because Eq. (6.1) constrains the external momenta, partial derivatives with respect to s, t, u must be represented by vector fields tangent to the massless kinematic manifold. A convenient choice is

$$\begin{aligned}\partial_s &= \frac{k_a \cdot \partial_{k_a} + k_b \cdot \partial_{k_b} - k_c \cdot \partial_{k_c}}{2s}, \\ \partial_t &= \frac{k_a \cdot \partial_{k_a} - k_b \cdot \partial_{k_b} + k_c \cdot \partial_{k_c}}{2t}, \\ \partial_u &= \frac{-k_a \cdot \partial_{k_a} + k_b \cdot \partial_{k_b} + k_c \cdot \partial_{k_c}}{2u}.\end{aligned}\tag{6.9}$$

After the homogeneous power of s is removed from each master, let $\mathbf{J}(x, y; \epsilon)$ denote the corresponding dimensionless original basis. Exact IBP reduction gives

$$d\mathbf{J} = [A_x(x, y, \epsilon) dx + A_y(x, y, \epsilon) dy] \mathbf{J}.\tag{6.10}$$

The system is flat,

$$\partial_x A_y - \partial_y A_x + A_y A_x - A_x A_y = 0,\tag{6.11}$$

and this identity holds as an exact rational function of x, y , and ϵ .

The non-Fuchsian double pole at $\Delta = 0$ is confined to the top block. It is removed by the near-identity Moser transformation

$$M_{\text{top}} = \mathbf{1}_2 + \frac{2\epsilon y(1-y)(2+3\epsilon)}{(1+\epsilon)\Delta} E_{21},\tag{6.12}$$

followed by a rational transformation of the full eight-dimensional basis. With $\mathbf{J} = T\mathbf{F}$, the result is the canonical system

$$d\mathbf{F} = \epsilon \sum_{a=1}^6 R_a d \log \phi_a \mathbf{F}, \quad \{\phi_a\} = \{x, 1-x, y, 1-y, 1-x-y, (1-x)^2 - y\},\tag{6.13}$$

where the residue matrices R_a are constant. This is the canonical $d \log$ form of Refs. [10, 11]. Substitution of the rational basis change back into Eq. (6.10) reproduces both differential matrices exactly, and the canonical connection is flat. In the chamber (6.3), every letter has a fixed sign along the transport path chosen below.

6.4 Nonuniform boundary and the independent corner period

The physical boundary $y \rightarrow 0^+$ is nonuniform. At the integral level it is the collinear configuration

$$k_c \longrightarrow x k_b,\tag{6.14}$$

and its cut-aware IBP reduction gives a separate seven-master boundary system. Near its endpoint $x = 0$, the physical Frobenius expansion contains three distinct modes,

$$x^0, \quad x^{-\epsilon}, \quad x^{-2\epsilon}.\tag{6.15}$$

The first is the hard region and the latter two are nonuniform collinear regions. Their lower-sector normalizations reduce to exact Beta and Gamma functions.

One further Frobenius coefficient is required to match these boundary modes to the canonical basis. Writing

$$\mathbf{B}_\lambda(x) = x^\lambda (b_0 + x b_1 + \mathcal{O}(x^2)), \quad \frac{d\mathbf{B}}{dx} = \left(\frac{A_{-1}}{x} + A_0 + \mathcal{O}(x) \right) \mathbf{B}, \quad (6.16)$$

with $\lambda \in \{0, -\epsilon, -2\epsilon\}$, the next coefficient is fixed by

$$[(\lambda + 1)\mathbf{1} - A_{-1}] b_1 = A_0 b_0. \quad (6.17)$$

The boundary-to-canonical matching matrix itself has a simple pole,

$$C(x) = \frac{C_{-1}}{x} + C_0 + \mathcal{O}(x), \quad (6.18)$$

so the finite canonical vector is

$$f_\lambda = C_0 b_0 + C_{-1} b_1, \quad (6.19)$$

not just $C_0 b_0$. This is an ordinary consequence of multiplying two Frobenius series, but it is essential here: retaining only the leading boundary vector would miss a finite contribution. The three matched vectors obey

$$\begin{aligned} R_x f^{(0)} &= 0, & (R_x + \mathbf{1}) f^{(1)} &= 0, & (R_x + 2\mathbf{1}) f^{(2)} &= 0, \\ R_y f^{(0)} &= 0, & R_y f^{(1)} &= 0, & R_y f^{(2)} &= 0, \end{aligned} \quad (6.20)$$

which supplies a regular tangential starting point at $y = 0$.

After the lower modes have been fixed, the triangular boundary equations leave one independent period in the top sector. In a boundary basis ordered so that its seventh element is the top integral, the last equation has the scalar form

$$\frac{dB_7}{dx} = \frac{D-6}{x-1} B_7 + c_1(x, \epsilon) B_1 + c_2(x, \epsilon) B_2 + c_3(x, \epsilon) B_3 + c_6(x, \epsilon) B_6, \quad (6.21)$$

where the c_i are exact rational functions. Its homogeneous solution is $(1-x)^{-2-2\epsilon}$; the inhomogeneous part is fixed by the lower modes, leaving one new corner constant. A manifestly real Euler representation for it is

$$\begin{aligned} r_{\text{top}}(\epsilon) &= - \frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\epsilon\Gamma(-2\epsilon)\Gamma(1-\epsilon)^3} \int_0^1 d\alpha d\beta d\gamma \alpha^{-1-\epsilon} \beta^{-1-\epsilon} \gamma^{-1-\epsilon} \\ &\quad \times (1-\alpha)^{-1-2\epsilon} (1-\beta)^{-1-2\epsilon} (1-\alpha\beta)^{2\epsilon} (1-\alpha\beta\gamma)^\epsilon. \end{aligned} \quad (6.22)$$

For $0 < \alpha, \beta, \gamma < 1$, every powered base in Eq. (6.22) is positive. The real branch is therefore fixed before the endpoint expansion. Resolving the correlated endpoints and performing the resulting linearly reducible coefficient integrals gives

$$\begin{aligned} r_{\text{top}}(\epsilon) &= - \frac{10}{\epsilon^3} + \frac{65}{\epsilon^2} - \frac{135}{\epsilon} + 90 - 8\zeta_3 \\ &\quad + \epsilon \left(52\zeta_3 - \frac{2\pi^4}{3} \right) + \epsilon^2 \left(\frac{13\pi^4}{3} - 108\zeta_3 - 288\zeta_5 \right) + \mathcal{O}(\epsilon^3). \end{aligned} \quad (6.23)$$

This is the only new period beyond the Beta/Gamma boundary data.

6.5 Ordered iterated-integral transport

The boundary is transported along the broken real path

$$(0, 0) \longrightarrow (x, 0) \longrightarrow (x, y). \quad (6.24)$$

The first segment has letters $\{0, 1\}$, while at fixed x the second has letters

$$\{0, 1, 1 - x, (1 - x)^2\}. \quad (6.25)$$

Equation (6.3) places the endpoint y below every nonzero letter on the second segment. Hence no singular letter is crossed, and the solution remains on the real physical branch.

We use the Chen–Goncharov iterated integral

$$\mathcal{G}(a_1, \dots, a_n; z) = \int_0^z \frac{dt}{t - a_1} \mathcal{G}(a_2, \dots, a_n; t), \quad \mathcal{G}(\cdot; z) = 1, \quad (6.26)$$

with tangential regularization at the lower endpoint; in particular, $\mathcal{G}(0, \dots, 0; z) = \log^n(z)/n!$ for a word of n zeros. The canonical solution is

$$\mathbf{F}(x, y) = U_y(x; y) U_x(x) \mathbf{F}_\partial, \quad U = P \exp \left[\epsilon \int \sum_a R_a d \log \phi_a \right]. \quad (6.27)$$

At weight n , a word $\mathcal{G}(a_1, \dots, a_n; z)$ multiplies the ordered matrix product $R_{a_1} \cdots R_{a_n}$. The residue matrices do not commute, so neither the word order nor the order $U_y U_x$ may be reversed. Although the target itself starts at ϵ^{-3} , the complete canonical boundary begins at ϵ^{-4} ; obtaining the target through ϵ^2 therefore requires transport through weight six.

6.6 Normalized analytic result

Let

$$\mathcal{N}_0(\epsilon) = s^{-1+2\epsilon} G_1(s, 0, 0; \epsilon). \quad (6.28)$$

The dimensionless phase-space and top masters are

$$\widehat{G}_1(x, y; \epsilon) = \frac{s^{-1+2\epsilon} G_1(s, t, u; \epsilon)}{\mathcal{N}_0(\epsilon)} = \Delta^{1-2\epsilon}, \quad \widehat{G}_7(x, y; \epsilon) = \frac{s^{3+2\epsilon} G_7(s, t, u; \epsilon)}{\mathcal{N}_0(\epsilon)}. \quad (6.29)$$

Consequently,

$$\frac{\widehat{G}_7}{\widehat{G}_1} = s^4 \frac{G_7}{G_1}, \quad (6.30)$$

which is independent of the common raw-measure convention.

The exact Laurent expansion is

$$\widehat{G}_7(x, y; \epsilon) = \sum_{n=-3}^2 \epsilon^n g_n(x, y) + \mathcal{O}(\epsilon^3), \quad g_{-4} = 0. \quad (6.31)$$

The first three nonzero coefficients are

$$g_{-3}(x, y) = -\frac{5}{\Delta^2}, \quad (6.32)$$

$$g_{-2}(x, y) = \frac{1}{\Delta^2} \left[\frac{65}{2} - 4\mathcal{G}(0; x) + 10\mathcal{G}(1; x) - 6\mathcal{G}(1; y) + 20\mathcal{G}(1 - x; y) \right], \quad (6.33)$$

and

$$\begin{aligned} g_{-1}(x, y) = \frac{1}{\Delta^2} \Big\{ & -\frac{135}{2} + \frac{2\pi^2}{3} + 26\mathcal{G}(0; x) - 65\mathcal{G}(1; x) + 2\mathcal{G}(0, 0; x) + 8\mathcal{G}(0, 1; x) \\ & + 4\mathcal{G}(1, 0; x) - 20\mathcal{G}(1, 1; x) + 39\mathcal{G}(1; y) - 6\mathcal{G}(0; x)\mathcal{G}(1; y) + 12\mathcal{G}(1; x)\mathcal{G}(1; y) \\ & - 130\mathcal{G}(1 - x; y) + 16\mathcal{G}(0; x)\mathcal{G}(1 - x; y) - 40\mathcal{G}(1; x)\mathcal{G}(1 - x; y) \\ & - 4\mathcal{G}(0; x)\mathcal{G}((1 - x)^2; y) - 12\mathcal{G}(0, 1; y) + 12\mathcal{G}(0, 1 - x; y) \\ & + 6\mathcal{G}(1, 1; y) + 12\mathcal{G}(1, 1 - x; y) + 24\mathcal{G}(1 - x, 1; y) - 80\mathcal{G}(1 - x, 1 - x; y) \\ & + 4\mathcal{G}((1 - x)^2, 1 - x; y) \Big\}. \end{aligned} \quad (6.34)$$

These equations display the complete pole part of the normalized top master. The finite, ϵ , and ϵ^2 coefficients are exact but substantially longer. After full expansion, g_0 , g_1 , and g_2 contain respectively 86, 313, and 1103 terms, built from 49, 153, and 473 distinct iterated integrals. The accompanying ancillary file `ancillary/NNLO_three_particle_cut_master.wl` contains every coefficient through ϵ^2 , with the path, branch convention, and normalization stated explicitly; it is not a table of numerical samples.

6.7 Independent fixed-point validation

At

$$s = 10, \quad x = \frac{3}{10}, \quad y = \frac{1}{5}, \quad (6.35)$$

define

$$R(\epsilon) = s^4 \frac{G_7}{G_1} = \frac{\hat{G}_7}{\hat{G}_1} = \sum_{n=-4}^2 r_n \epsilon^n + \mathcal{O}(\epsilon^3). \quad (6.36)$$

The analytic GPL expression gives the coefficients in Table 1. An independently generated AMFLOW evaluation of the same cut family agrees coefficient by coefficient at relative tolerance 10^{-12} .

Power n	r_n
-4	0
-3	-40
-2	282.32024127181498559344
-1	-662.90998166471340573018
0	418.83703822298626042003
1	-12.21808964614470122794
2	-32.81177029956454278367

Table 1. Independent fixed-point validation of the exact analytic top master. The displayed numbers evaluate the GPL solution; they do not define or fit it.

For reproducibility, the symbolic derivation used the FEYNCalc objects `FCTopology` and `GLI`, together with `FCLoopGLIDifferentiate`, for the family and invariant derivatives

[12]; KIRA through `FeynKiraReduce` for exact cut-aware IBP reduction [6, 7]; and the CANONICA functions `CheckEpsForm` and `CalculateDlogForm` for the canonical system. The ordered Chen transport through weight six was generated with LIBRA’s `PexpExpansion` [9]. The same migration was applied to the f228 and d099 families; all three resulting analytic associations are exactly identical to the corresponding pre-migration expressions, and the 83bb transport satisfies both differential equations through the required order. The three-mode 83bb boundary certificate is stored with the differential-equation source rather than inferred by LIBRA. The positive Euler period was integrated with SUBTROPICA’s `STIntegrate` and exact HYPERINT-compatible hyperlogarithmic functions [8, 13, 14]. Finally, AMFLOW’s `SolveIntegrals` was used only for the independent numerical check in Table 1 [15].

7 BMHV sectors and reconstruction

For transverse polarization, the BMHV split is [16]

$$g_D^{\mu\nu} = \bar{g}^{\mu\nu} + \hat{g}^{\mu\nu}, \quad \ell^\mu = \bar{\ell}^\mu + \hat{\ell}^\mu, \quad \mu_\epsilon^2 = -\hat{\ell}^2. \quad (7.1)$$

Declared external vectors have vanishing evanescent components, but powers of μ_ϵ^2 may remain in a loop numerator. Such powers are retained in the post-IBP manifest and converted with normalized dimension shifts of the Tarasov type [17]. Both the dimension-dependent reduction coefficient and the shifted master are evaluated in the shifted dimension; coefficients inherited from the original trace remain in $D = 4 - 2\epsilon$.

After every master has been evaluated to sufficient depth, the exact IBP coefficients and master series are multiplied and summed. Endpoint analysis is performed only on this reconstructed invariant result. For $w \in [0, 1]$,

$$(1-w)^{-1-a\epsilon} = -\frac{1}{a\epsilon}\delta(1-w) + \sum_{m=0}^{\infty} \frac{(-a\epsilon)^m}{m!} \mathcal{D}_m(w), \quad \mathcal{D}_m(w) = \left[\frac{\ln^m(1-w)}{1-w} \right]_+. \quad (7.2)$$

The coefficient of each endpoint branch is extracted from the exact analytic expression before the ϵ expansion. This retains both $(1-w)^{-1-\epsilon}$ and $(1-w)^{-1-2\epsilon}$ when both occur.

The real contribution is then combined with the virtual correction, renormalization, and collinear counterterms in one convention. Reverse unitarity by itself neither supplies those terms nor guarantees cancellation of all poles or the final $\delta(1-w)$ coefficient.

Part II

Implementation record

8 Software architecture and data ownership

The current implementation uses FEYNARTS for diagram generation [18], FEYNALC for algebra and topology objects [12], FEYNHELPERS for interfaces to reduction tools [19, 20],

KIRA for IBP reduction, SUBTROPICA for analytic parametric integration, and LIBRA for exact path-ordered transport of canonical differential systems [9]. POLYLOGTOOLS and HPL are restricted to branch-controlled polylogarithmic cleanup after the physical branch has been fixed [21, 22]. They are not allowed to supply a boundary condition or silently analytically continue an expression. Table 2 states the active ownership boundaries.

9 Collinear-factorization front end

The front end is driven by an explicit association. Incoming and outgoing partons have aligned records for momentum, momentum fraction, hadron momentum, light-cone directions, longitudinal spin, and transverse spin. Independent loop orders and diagram indices are supplied for the amplitude and its conjugate; ghost fields are not excluded by design when a loop calculation requires them.

The public call has the form

```
frontEnd = CollinearFactorize[<|
  "Diagrams" -> diagrams,
  "Partons" -> partons,
  "PartonMomentum" -> partonMomentum,
  "MomentumFraction" -> momentumFraction,
  "HadronMomentum" -> hadronMomentum,
  "HadronLongDirection" -> hadronLongDirection,
  "HadronDualDirection" -> hadronDualDirection,
  "HadronLongSpin" -> hadronLongSpin,
  "HadronTransSpin" -> hadronTransSpin,
  "PhaseSpaceMomentum" -> phaseSpaceMomentum,
  "PartonIntegrated" -> partonIntegrated,
  "DiagramIndex" -> diagramIndex,
  "Assumptions" -> assumptions
|>];
```

It returns

$$\{d\Xi, \mathcal{P}, d\Pi_D, \mathcal{N}, \mathcal{R}\}, \quad (9.1)$$

where $d\Xi$ is the convolution-fraction measure, $\mathcal{P}$ is a loop-independent prefactor, $d\Pi_D$ is the remaining integration measure, $\mathcal{N}$ is a propagator-free numerator, and $\mathcal{R}$ contains ordinary and cut propagators. This separation is an enforced contract, not display formatting.

The front end performs the following operations in order:

1. generate and select the ordered amplitude pair;
2. perform spin and vector-polarization sums;
3. insert the collinear density matrices on hadron-associated legs;
4. apply declared massless, distribution-zero, and evanescent-zero data;
5. eliminate only the explicitly declared integrated momentum;

Component	Responsibility
FeynNLO package	Process records, light-cone basis checks, leading-twist density matrices, phase-space construction, cut representation, common-factor separation, and the five-field collinear-factorization output.
DY_FeynArts_Born.nb	Minimal generated-amplitude test of the density-projector front end.
FeynNLO_real_test.nb	Real-emission test of the full collinear front-end contract and literal denominator extraction.
ReverseUnitarity/	Cut-preserving partial fractions, invariant family discovery, Kira input, reduction import, and regression against the established NLO result.
EvaluateMasters/	Frozen post-IBP manifests, analytic master evaluation, exact audits, independent reference checks, canonical generated outputs, and adapters that enforce the exact-package contracts.
pp-hX_Real.nb, pp-hX_Virtual.nb, pp-hX_Renormalized.nb	Established real, virtual, and combined NLO calculations used as independent physics benchmarks.

Table 2. Ownership boundaries in the current project.

6. separate safe common factors without extracting scalar products, denominators, cuts, or color objects;
7. return Eq. (9.1) and reject residual mixed ownership.

No v, w variables, angular coordinates, expected fraction list, or master library enters this stage.

10 Reduction and agent protocol

For every fraction generated from $\mathcal{R}$, the reduction layer stores the literal denominator list, mandatory cuts, completed `FCTopology`, GLI targets, and imported Kira rules. A family is accepted only if `FCLoopValidTopologyQ` is true and `FCLoopBasisIncompleteQ` is false. Every designated cut position must remain positive in targets and masters.

In the implementation, an algebraic match between completed topology records is used only to generate a candidate transformation. The candidate is accepted only after the affine map, strict propagator permutation, exponent permutation, cut-set map, energy-orientation condition, ordinary causal prescriptions, and available physical metadata have all been checked as in Sec. 4. In particular, the original $i0$ and stored η -sign information are retained while the denominator polynomials are compared and are tested again after the algebraic map has been reconstructed. For the present NLO real-emission sample, the 39 completed records form eight cut-aware equivalence classes, giving 31 maps from non-representative records to their class representatives. The partition was unchanged under 20 random orderings of the input, under independent relabellings of the propagator slots of all 39 records with their decorations transported consistently, and for 40 generated affine reroutings of the eight representatives. Generated two- and three-loop orbits were also recovered. Deliberate mutations of the cut direction, cut set, causal prescription, η sign, measurement role, and loop-map Jacobian were rejected, while 800 random applications of the accepted index maps preserved every exponent under the prescribed permutation. As an independent reduction-level check, all 39 families were submitted to Kira with their cut sets declared but without supplying the eight FACET classes; the same denominator-and-cut-set partition was obtained and no requested target remained unreduced. The Kira comparison corroborates the scalar reduction layer, whereas the energy-orientation, causal, and measurement conditions are established by the decorated-topology checks. These tests validate the present 39-record data set and the generated examples through three loops, but do not constitute a completeness proof for arbitrary future topologies.

The post-IBP export to `EvaluateMasters` is a compact association:

```
<|
  "Family" -> topology,
  "Powers" -> powers,
  "CutPositions" -> cuts,
  "Prescriptions" -> prescriptions,
  "Coefficient" -> coefficient,
  "PhysicalAssumptions" -> physicalRegion,
```

```

"RequestedEpsilonOrder" -> order,
"BMHVSector" -> sector
|>

```

The evaluator first audits this manifest. It then derives or loads the exact parameterization, proves its identities, evaluates only the required analytic shapes, canonicalizes the result, and writes the output atomically after every required check passes.

The current NLO evaluator derives the cut invariant, ordinary-line scales, and cross ratio directly from saved `FCTopology` objects. It reduces six physical instances to two generic analytic functions. In the project normalization they are

$$B(\epsilon) = \frac{2\pi}{1-2\epsilon}, \quad (10.1)$$

$$T(z, \epsilon) = -\frac{\pi}{\epsilon\Lambda_1\Lambda_2} {}_2F_1(1, 1; 1-\epsilon; z). \quad (10.2)$$

The physical normalization multiplying these reduced functions is

$$\frac{(4\pi)^\epsilon \Gamma(1-\epsilon)}{16\pi^2 \Gamma(1-2\epsilon)} W^{-\epsilon}, \quad W = s + t + u. \quad (10.3)$$

For every family, the evaluator proves $0 < z < 1$ from the declared physical region rather than using a numerical witness.

10.1 NNLO structural inventory

The saved one-flavor double-real process $ud \rightarrow udgg$ contains 36 amplitudes and 666 unordered amplitude pairs. Literal denominator data are classified under only one equivalence: a simultaneous interchange of the two unresolved gluons on both sides of the cut. This produces 68 denominator-class representatives while retaining the relative orientation of the amplitude and its conjugate.

The representative reduction sweep is complete. Table 3 records the resulting structural inventory.

Object	Count
Denominator representatives	68
Nonzero / zero reductions	59 / 9
Active family-labelled masters	4,242
Canonical support shapes	157
Ordinary / doubled / tripled cut shapes	120 / 36 / 1
Shapes containing numerator factors	30
Maximum ordinary denominator factors	6

Table 3. Completed NNLO denominator-representative inventory as of 2026-07-29.

Each shape is identified by a stable SHA-256 key constructed from its canonical cut powers, numerator powers, and ordinary denominator data. Ordinal display identifiers are not persistent under an inventory rebuild. The previously validated pair $\{19, 24\}$ is

imported into the same pair-record schema as the other 58 nonzero reductions before coverage is counted.

The distinction among the numbers in Table 3 is essential. A family-labelled master, a support shape after a proved variable change, and an independent analytic function are different objects. The 157 shapes exhaust the 68 denominator representatives. They are not yet a proved full-process target union because different numerators among the 666 pairs may activate additional integrals. Closure requires either an all-pair target-union audit or a proved numerator-rank seed bound for every family.

The complete regression is launched under Linux or WSL with

```
cd EvaluateMasters
bash validation/run_all.sh
```

It parses all sources, evaluates NLO and the representative NNLO fixture in fresh kernels, runs the independent checks, audits the generated associations, and repeats the complete sequence. Canonical output must be byte identical; runtime measurements are excluded from the saved associations.

11 Validation record

11.1 Verified coverage

Table 4 records the present analytic and structural coverage. A resolved fixed-point numerical integral is deliberately excluded from the analytic rows.

Scope	Status	Meaning
NLO analytic masters	Pass, six physical instances	One common two-cut bubble and five top masters are symbolic in s, t, u, ϵ and reduce to Eqs. (10.1) and (10.2). Exact branch, differential-equation, and boundary checks pass.
Corrected NNLO representative pair	Pass, seven fixed-point shapes	Exact Laurent data were obtained at one rational kinematic point for C , C/R , $C/(AR)$, $C/(AX)$, C_2 , C_2/R , and $C_2/(AR)$. This is not symbolic multiscale coverage.
NNLO denominator inventory	Pass, 68/68 classes	The sweep has 59 nonzero and nine zero reductions. Its 4,242 active family labels canonicalize to 157 support shapes. Numerator completeness over all 666 pairs is not yet proved.
Hard-master stress campaign	Pass, three of twelve complete	The 83bb, f228, and d099 families contain 8, 4, and 5 masters, respectively. All 17 masters are exact multiscale GPL expansions through ϵ^2 ; one target in each family passes an independent coefficient-level AMFlow comparison. The other nine stress families remain unresolved.

Table 4. Verified calculation coverage as of 2026-07-31.

The seven NNLO shapes belong to the UU $ud \rightarrow udgg$ pair {19, 24}. A fresh Kira run with the corrected numerator

$$\frac{s}{2}N_{001} + N_{111} \quad (11.1)$$

It reproduced 835 fractions, six topologies, 21 family labels, and 19 active labels. Those active labels map to the seven evaluated shapes. One exact identity, Eq. (5.3), bounds their analytic rank by six. The seven fixed-point formulas therefore establish the corrected pair fixture, but neither symbolic multiscale coverage nor the full NNLO master count.

The larger inventory in Table 3 is a separate structural result. A later stress campaign selected twelve physically inequivalent difficult orbits. Three are now complete analytic families: 83bb has eight masters, f228 has four, and d099 has five. Their exact differential systems, analytic boundary constants, and GPL transports are available through ϵ^2 . The coupled 83bb derivation is presented in Sec. 6. These 17 masters are substantial tests of the analytic agent, but the remaining nine stress families and the numerator-complete NNLO amplitude remain open; they therefore do not constitute a full NNLO result.

11.2 AMFlow verification

AMFlow supplies the principal independent numerical verification of the analytic masters [15]. For every comparison it receives the same family, denominator powers, selected cut slots, ordinary-line prescriptions, and physical kinematics as the analytic calculation. It

therefore checks the normalization and causal definition of a particular cut master, rather than a global discontinuity of its parent graph.

At NLO, all six physical master instances were evaluated independently at

$$(s, t, u, \epsilon) = (10, -3, -2, -1/10). \quad (11.2)$$

Each agrees with the symbolic result within relative tolerance 10^{-15} . At NNLO, all seven fixed-point shapes have an independent reference comparison. The phase-space volume C and the doubled-cut shape C_2 satisfy exact normalization and ratio checks. The five remaining nontrivial ratios have independent full- ϵ quadrature checks, including a separate convergent Appell representation for $C_2/(AR)$.

The three completed hard families have a deeper independent test at the common physical point

$$(s, t, u) = (10, -3, -2), \quad (x, y) = (3/10, 1/5). \quad (11.3)$$

For one target in each of 83bb, f228, and d099, the exact GPL solution and AMFLOW agree coefficient by coefficient from the leading pole through ϵ^2 at tolerance 10^{-12} . Table 1 displays the hardest case. No numerical coefficient enters a differential equation, boundary constant, or analytic reconstruction.

For these families the canonical transport is now generated by LIBRA, while the physical boundary data are derived and audited separately. Rebuilding 83bb, f228, and d099 with this package-assisted transport gives exact *Mathematica* equality with the earlier manually enumerated Chen series. This is a migration check on the analytic representation, independent of the fixed-point AMFLOW comparison.

The AMFLOW manifests and their generation scripts are stored separately from the analytic evaluator and are hash-audited before comparison. Standard AMFLOW output at fixed kinematics is numerical. It can reject an analytic formula, reveal a missing normalization, or constrain a diagnostic boundary, but it does not define the delivered analytic master.

11.3 Other independent checks

The complete regression recomputes the NLO and representative NNLO analytic associations and their independent comparisons in fresh kernels, then requires all four canonical outputs to be byte identical. The latest sequential run passes in 91.8 seconds. Separately, the NNLO inventory audit passes all 12 structural checks, including stable-key uniqueness, imported-pair coverage, cut-power counts, and the recorded completeness boundary.

The established NLO reverse-unitarity calculation supplies an independent end-to-end benchmark: all 25 UU and LL diagram pairs were fractioned automatically, reduced through five families, reconstructed, and compared with the stored direct integration. The master-identity and high-precision diagram comparisons pass. This benchmark validates the invariant reduction; it is not used to define Eqs. (10.1) and (10.2).

12 Fail-fast checks and current boundary

The main checks are summarized in Table 5. A failed required check stops the stage and prevents output from being written.

Table 5: Required checks in the current workflow.

Stage	Required invariant
Process record	Side lengths agree; momenta identify unique legs; diagram indices and loop orders are valid; missing hadron associations are explicit.
Collinear front end	All expected traces and polarization sums close; the five returned fields satisfy Eq. (9.1); no eliminated momentum survives.
Partial fractions	The sum reproduces the literal rational expression and every term retains every mandatory cut.
Integral family	The topology is valid and complete; cut slots are sorted, unique, and in range; no target pinches a cut.
Topology equivalence	A unit-measure affine loop transformation induces a strict permutation of the generalized propagator slots; the external kinematics, cut subset and energy orientations, ordinary $i0$ prescriptions, and all available line-level physical or measurement labels are preserved under the same permutation.
Kira import	Reduction files exist; all residual GLIs are declared masters; family, powers, cuts, and prescriptions survive translation.
Master manifest	Exact coefficients, physical region, BMHV sector, and required Laurent depth are present before evaluation begins.
Parameterization	Jacobian, inverse map, integrand identity, phase-space normalization, and positivity of every powered base are proved exactly.
Analytic result	No unresolved wrapper, conditional result, machine real, or unproved branch remains; exact identities and required Laurent coverage pass.
Independent validation	Numerical checks use a separately generated reference and do not enter the analytic output. A repeated run is byte identical.
Reconstruction	Every master is replaced, all endpoint branches are retained, and the result contains no unresolved cut-integral or reduction object.

The present milestone is a symbolic NLO master stage, a separately verified NLO hard-function calculation, one corrected NNLO pair evaluated at a fixed physical point, a

complete denominator-representative inventory for the saved NNLO double-real process, and three exact multiscale hard families through ϵ^2 . The master agent is not yet connected to every diagram automatically, and the TT channel still requires full end-to-end BMHV validation. The numerator target union and the remaining nine stress families remain open. The current inventory and the 83bb, f228, and d099 solutions do not constitute a general analytic NNLO result.

Measured observables and event shapes may introduce additional cut constraints or measurement denominators. Those objects must enter the integral manifest explicitly. They cannot be hidden in a post-processing rule or inferred from a global discontinuity.

13 Alternative workflows explored

This section records routes that remain useful as checks or fallbacks but are not the main workflow described above.

One global discontinuity of an uncut forward integral. For a graph with several allowed cuts in the same channel,

$$\text{Disc}_s I = \sum_{C \in \mathcal{C}_s} \mathcal{N}_C \text{Cut}_C I. \quad (13.1)$$

A single discontinuity therefore returns a sum of physical cuts and does not generally isolate the measured final state [23]. It also does not by itself encode every positive-energy theta function. This route is valid only after proving that the desired cut is the unique contribution, or after constructing and subtracting all other cuts analytically. It is not used as a general replacement for the explicit cut set in Eq. (3.3).

Differential equations with boundary data. Equation (5.4) is the preferred analytic fallback when direct parametric integration fails. Its limitation is not the equation but the boundary problem. A boundary derived from an exact volume, simpler cut, controlled limit, regularity, or symmetry is acceptable. A fitted high-precision constant is useful for discovery and verification but does not meet the analytic acceptance criterion. Automation is therefore conditional on proving the boundary data.

Direct cut and residue representations. Loop-tree and residue formulations can expose physical integration domains without treating the cut as a global graph discontinuity [24]. They may become useful for constructing exact parameterizations, especially when several cuts coexist. They still require a proof of energy flow, normalization, and branch choice and have not replaced the current post-IBP parameterization stage.

Pre-tabulated angular masters. An earlier NLO implementation introduced a center-of-mass frame after Kira and reconstructed every master from a small analytic angular library. This route is fast and remains an independent NLO benchmark. It is no longer the main design because extending a manually curated library to new masses, measurements, and loop orders moves process knowledge into the wrong layer. The current agent instead derives and verifies each required parametric density from the frozen post-IBP manifest.

One-loop closed-form integral libraries. Closed-form one-loop tools remain useful for virtual corrections and simple cross-checks. Their integral class and available ϵ depth are too limited to define the general real-emission or multiloop master-evaluation strategy.

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